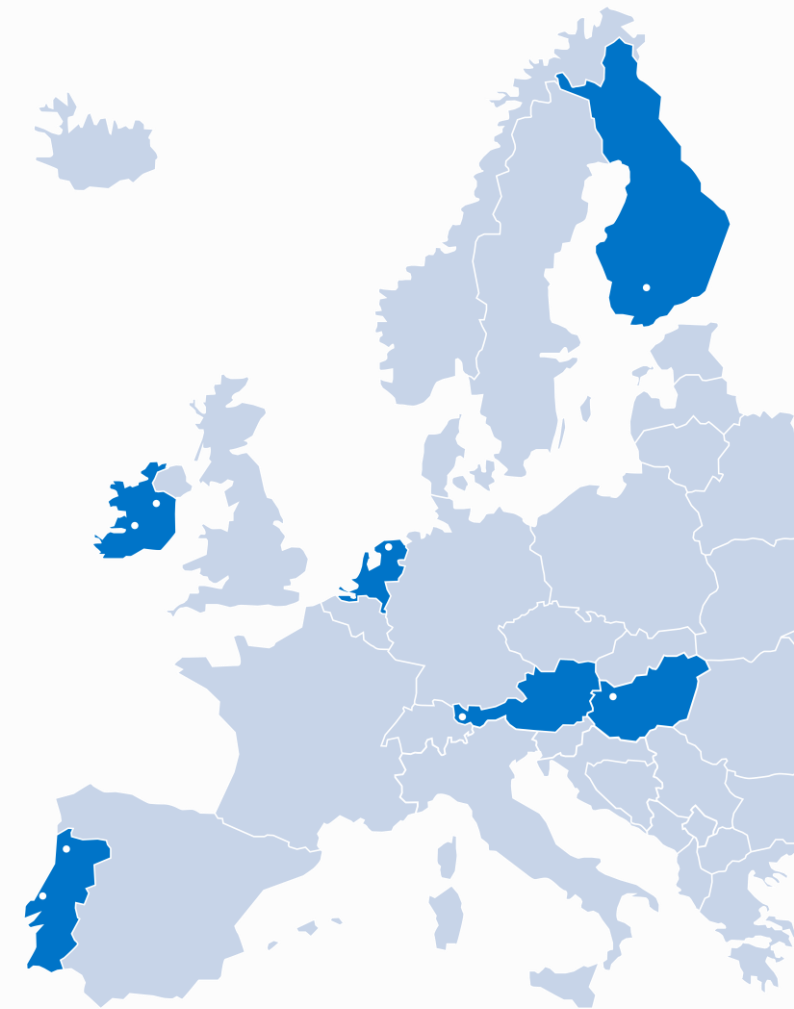


BIP – “HACK YOUR DEVICE” FUNDAMENTALS OF CYBERSECURITY

SETTING UP YOUR HACKING ENVIRONMENT





AGENDA

- *Hacking environment*
- *Setting up a virtual machine with Kali Linux OS*
- *Hacking tools*
- *Anonymization*
- *Exploration*



HACKING ENVIRONMENT

- Hardware (PC, switch, router, network cables, network cards (wired/wireless), ...)
- Virtual Machines (VMware Workstation Player, Oracle VirtualBox, ...)
- Containers (Docker, Kubernetes, ...)
- Scripting (python, bash, ...)
- VM Windows OS
 - <https://developer.microsoft.com/en-us/windows/downloads/virtual-machines>
 - <https://www.microsoft.com/en-us/evalcenter/evaluate-windows-10-enterprise>
 - <https://www.microsoft.com/en-us/evalcenter/evaluate-windows-11-enterprise>

HACKING ENVIRONMENT

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HACKING ENVIRONMENT



HACKING ENVIRONMENT

- *Linux OS solutions*

- Kali Linux
- Parrot OS
- BackBox
- OWASP SamuraiWTF
- Wifislax Operating System
- ArchStrike
- Network Security Toolkit
- Attify OS: specifically for IoT

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HACKING ENVIRONMENT

- *Windows OS solutions*
 - commando-vm
 - Pentestbox





SETTING UP A VIRTUAL MACHINE WITH KALI LINUX

1. Install a desktop application that lets you run a virtual machine on a Windows/Linux/Mac PC like VMware Workstation or VirtualBox.
2. Get an Installer Image or a Pre-built Virtual Machine of Kali Linux.
3. Open and run a Kali Linux virtual machine.
4. Use the default credentials "kali/kali".

SCAN ME



SETTING UP A VIRTUAL MACHINE WITH KALI LINUX (COMMON PROBLEMS)

1. Confirm that your hardware supports virtualization
2. Enable Virtualization in your BIOS
3. Sometimes Hyper-V needs to be disabled (on Windows)
4. Differences 64-bit virtual machine on a 32-bit computer (doesn't work)
5. Enough disk space
6. Good resources for networking and VirtualBox

[VirtualBox Network Settings: All You Need to Know \(nakivo.com\)](https://www.nakivo.com/virtualbox-network-settings/)



HACKING TOOLS

- Footprinting (Spiderfoot, osintframework.com, ...)
- Sniffers (wireshark, tcpdump, ...)
- Pwd “crackers” (John the Ripper, hashcat, ...)
- Fuzzing (Burp Suite, Sulley, ...)
- Disassemblers e Debuggers (Ghidra, OllyDbg, ...)
- Others: <https://linuxsecurity.expert/security-tools>

ANONYMIZATION

Used to protect the identification, information and actions during hacking activities.

- *MAC address level*

- macchanger (-s | -l | -r | -m | -p)
- Technitium MAC Address Changer

- *IP address level*

- Proxy
 - hidemyass.com
 - anonymouse.org
 - Web browser: Tor / extensions / add-ons / anonymoX

ANONYMIZATION

- *IP address level*

- VPN

- freevpn.me
 - torguard.net
 - spotflux.com
 - hotspotshield.com
 - ultrasurf.us

ANONYMIZATION

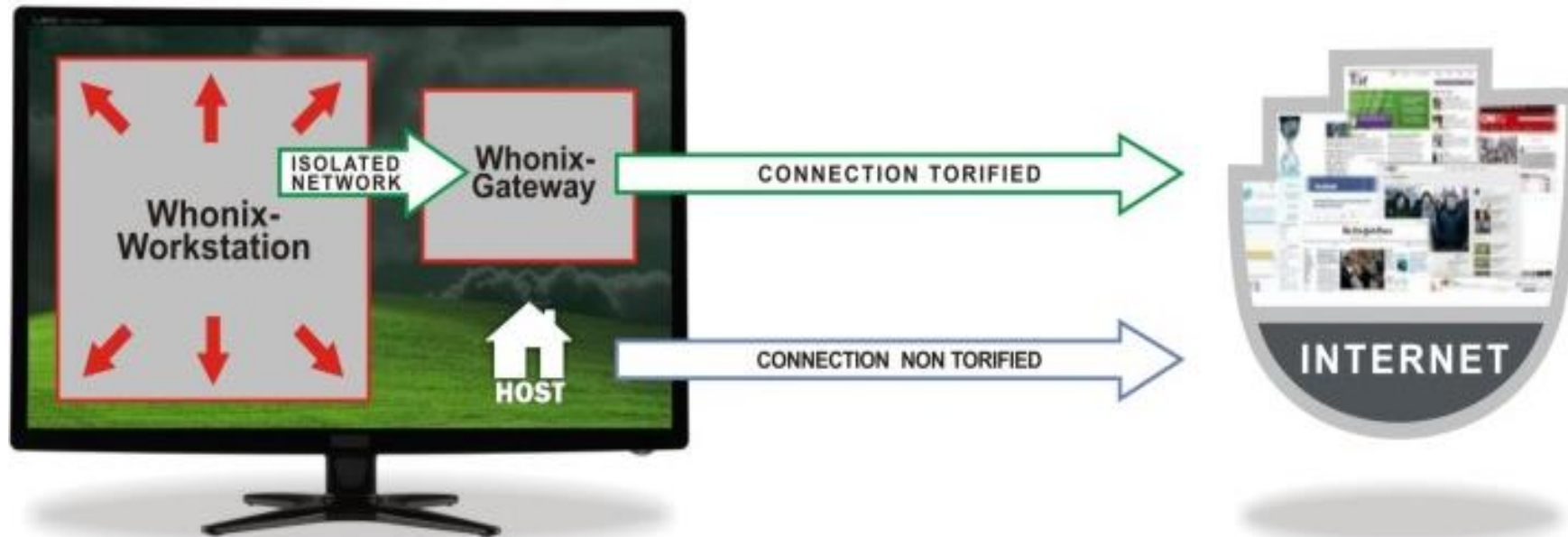
- *Kali Linux*
 - kali-anonsurf
 - anonsurf help / start / change / myip / stop
 - kalitorify
 - kali-anonymous
 - 4nonimizer

ANONYMIZATION

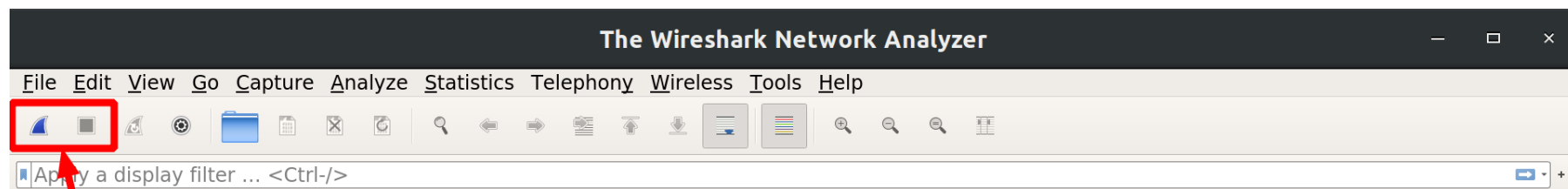
- *Temporary e-mail box/address*
 - temp-mail.org
 - tuta.com
 - emailondeck.com
 - tempail.com
 - throwawaymail.com

ANONYMIZATION

- *Whonix solution*
 - 2 VM = 1 workstation + 1 Tor “gateway”



WIRESHARK - CAPTURE WIFI TRAFFIC



Start/Stop Buttons

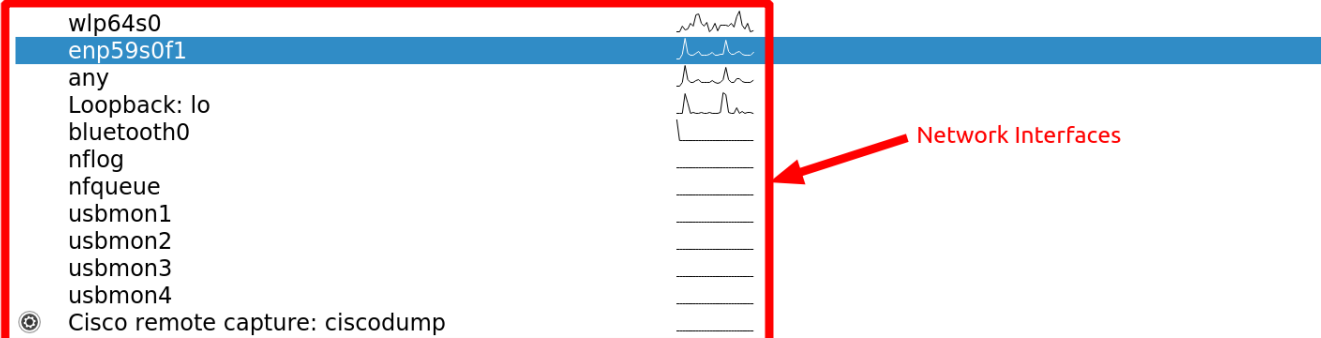
Welcome to Wireshark

Capture

...using this filter:

Capture Filter

All interfaces shown



Network Interfaces

Learn

[User's Guide](#) · [Wiki](#) · [Questions and Answers](#) · [Mailing Lists](#)

You are running Wireshark 3.2.5 (Git v3.2.5 packaged as 3.2.5-1~ubuntu18.04.0+wiresharkdevstable1).

WIRESHARK - FILTERS

- IP address:
 - ip.addr == 192.168.0.5
 - !(ip.addr == 192.168.0.0/24)
- Protocol:
 - tcp
 - tcp.port == 80 || udp.port == 80
 - http
 - ip.src == 10.43.54.65 and tcp.dstport == 25

ip.src == 23.217.250.58 The background is green which means the filter is correct

No.	Source	Destination	Protocol	Info
5	23.217.250.58	192.168.1.4	TCP	443 → 50526 [ACK] Seq=1
105	23.217.250.58	192.168.1.4	TCP	[TCP Keep-Alive ACK] 443
220	23.217.250.58	192.168.1.4	TCP	[TCP Keep-Alive ACK] 443
300	23.217.250.58	192.168.1.4	TCP	[TCP Keep-Alive ACK] 443
305	23.217.250.58	192.168.1.4	TLSv1.2	Application Data
306	23.217.250.58	192.168.1.4	TCP	443 → 50526 [FIN, ACK] Seq=1
349	23.217.250.58	192.168.1.4	TCP	[TCP Keep-Alive ACK] 443
460	23.217.250.58	192.168.1.4	TCP	443 → 50526 [RST] Seq=20
5530	23.217.250.58	192.168.1.4	TCP	443 → 50678 [SYN, ACK] Seq=1
5536	23.217.250.58	192.168.1.4	TCP	443 → 50678 [ACK] Seq=1
5537	23.217.250.58	192.168.1.4	TLSv1.3	Server Hello, Change Ciphers
5538	23.217.250.58	192.168.1.4	TCP	443 → 50678 [PSH, ACK] Seq=1
5539	23.217.250.58	192.168.1.4	TCP	443 → 50678 [PSH, ACK] Seq=1
5541	23.217.250.58	192.168.1.4	TCP	443 → 50678 [ACK] Seq=40
5543	23.217.250.58	192.168.1.4	TLSv1.3	Certificate, Certificate

WIRESHARK - FILTERS

- `ip.addr == 10.0.0.1`: Show all traffic with 10.0.0.1 as either source or destination.
- `ip.addr == 10.0.0.0/24`: Show all traffic to and from any address in 10.0.0.0/24.
- `ip.src == 10.0.0.1 && ip.dst == 10.0.0.2`: Show all traffic from 10.0.0.1 to 10.0.0.2.
- `!(ip.addr == 10.0.0.1)`: Exclude all traffic to or from 10.0.0.1.
- `icmp.type == 3`: Show ICMP "destination unreachable" packets.
- `tcp or udp`: Show TCP or UDP traffic.
- `tcp.port == 80`: Show TCP traffic with port 80.
- `tcp.srcport < 1000`: Show TCP traffic with source port range.
- `http or dns`: Show all HTTP or DNS traffic.
- `tcp.flags.syn == 1`: Show TCP packets with SYN flag set.
- `tcp.flags == 0x012`: Show TCP packets with both SYN and ACK flags set.
- `tcp.analysis.retransmission`: Show all retransmitted TCP packets.
- `http.request.method == "GET"`: Show TCP packets associated with HTTP GET.
- `http.response.code == 404`: Show packets associated with HTTP 404 response.
- `http.host == "www.abc.com"`: Show HTTP traffic matching the Host header field.
- `tls.handshake`: Show only TLS handshake packets.
- `tls.handshake.type == 1`: Show client Hello packet during TLS handshake.
- `dhcp and ip.addr == 10.0.0.0/24`: Show DHCP traffic for 10.0.0.0/24 subnet.
- `dhcp.hw.mac_addr == 00:11:22:33:44:55`: Show DHCP packets for client MAC address.
- `dns.resp.name == cnn.com`: Show DNS responses with name field of "cnn.com".
- `frame contains keyword`: Show all packets that contain the word "keyword".
- `frame.len > 1000`: Show all packets with total length larger than 1000 bytes.
- `eth.addr == 00:11:22:33:44:55`: Show all traffic to or from the specified MAC address.
- `eth[0x47:2] == 01:80`: Match Ethernet frames with 2 bytes at offset 0x47 == 01:80.
- `!(arp or icmp or stp)`: Filter out background traffic from ARP, ICMP, and STP.
- `vlan.id == 100`: Show packets with VLAN ID 100.

WIRESHARK – PACKET DETAILS

17	0.232621	200.121.1.131	172.16.0.122	TCP	1454	10554 → 80 [ACK]	Seq=12601 Ack=1 Win=65535 Len=1400 [TCP segment of ...
18	0.232629	172.16.0.122	200.121.1.131	TCP	54	80 → 10554 [ACK]	Seq=1 Ack=14001 Win=63000 Len=0
19	0.258365	200.121.1.131	172.16.0.122	TCP	1454	10554 → 80 [ACK]	Seq=14001 Ack=1 Win=65535 Len=1400 [TCP segment of ...
20	0.258373	172.16.0.122	200.121.1.131	TCP	54	80 → 10554 [ACK]	Seq=1 Ack=15401 Win=63000 Len=0

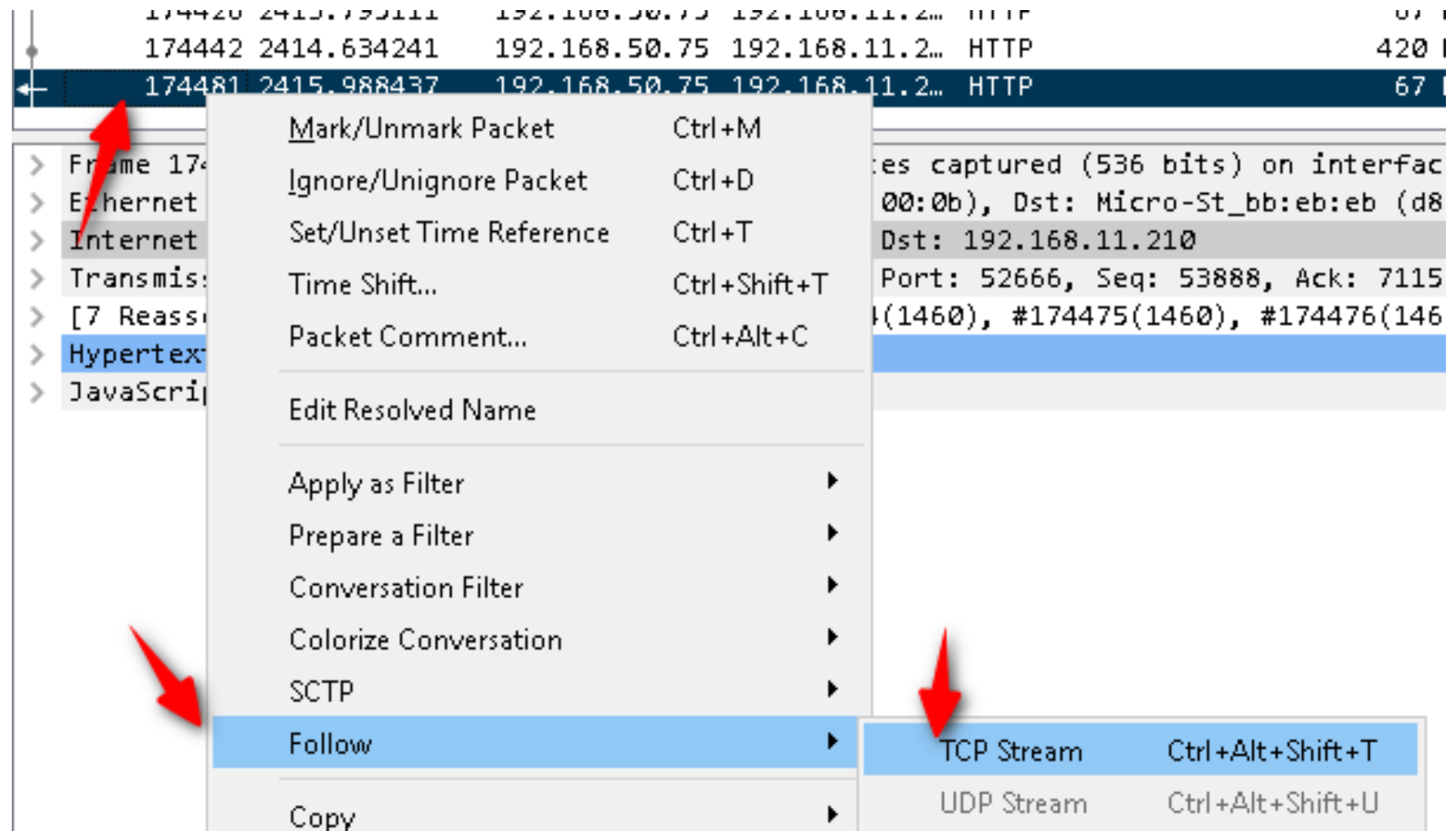
> Frame 15: 1454 bytes on wire (11632 bits), 1454 bytes captured (11632 bits)
 > Ethernet II, Src: Vmware_c0:00:01 (00:50:56:c0:00:01), Dst: Vmware_42:12:13 (00:0c:29:42:12:13)
 > Internet Protocol Version 4, Src: 200.121.1.131, Dst: 172.16.0.122
 ✓ Transmission Control Protocol, Src Port: 10554, Dst Port: 80, Seq: 11201, Ack: 1, Len: 1400

Source Port: 10554
 Destination Port: 80
 [Stream index: 0]
 [TCP Segment Len: 1400]
 Sequence number: 11201 (relative sequence number)
 [Next sequence number: 12601 (relative sequence number)]
 Acknowledgment number: 1 (relative ack number)
 0101 = Header Length: 20 bytes (5)

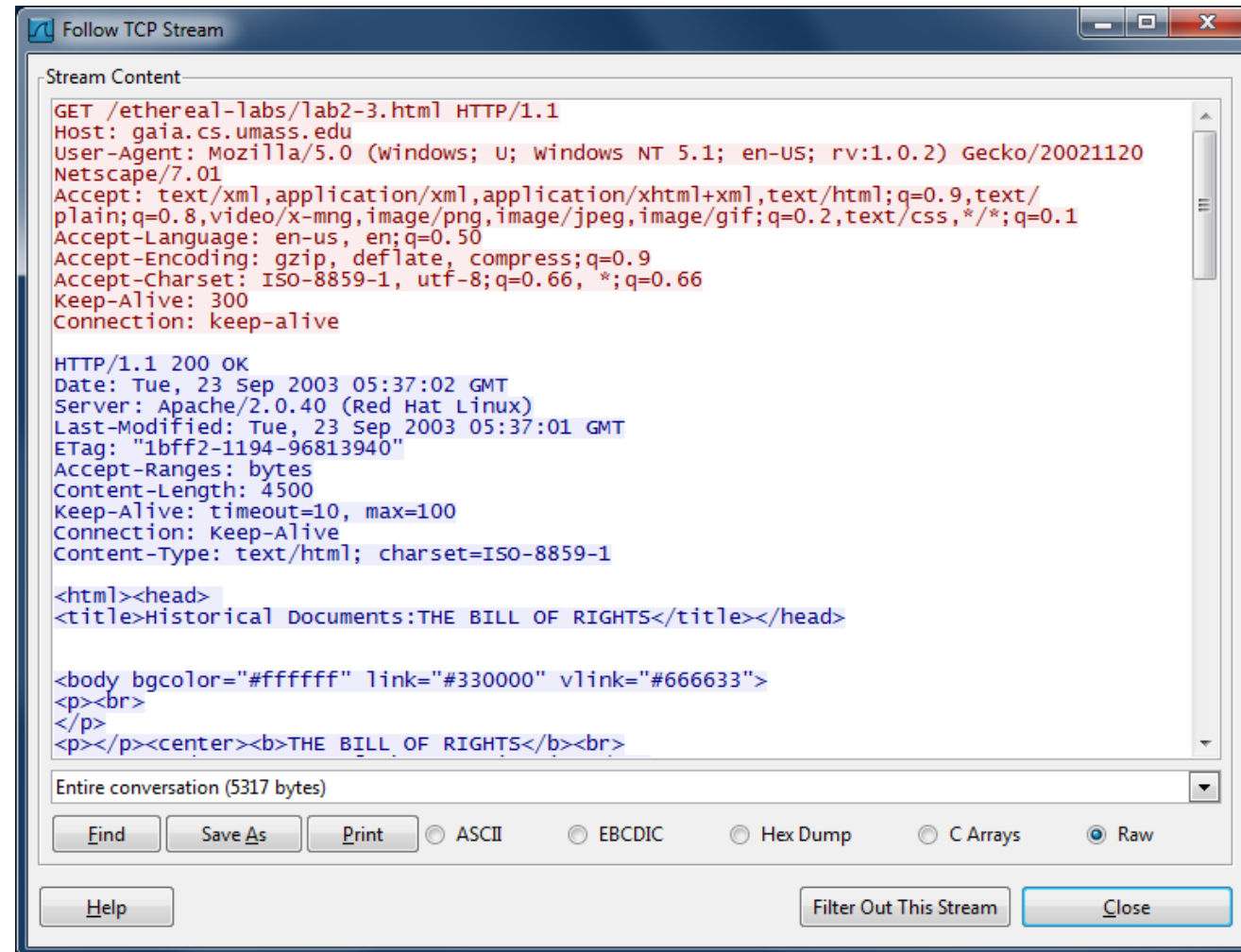
0020	00 7a 29 3a 00 50 a7 5c 30 08 e2 e2 ee bf 50 10	·z)·P·\ 0.....P·
0030	ff ff bc 5e 00 00 42 4f 78 42 56 35 6a 45 52 52	...^...BO xBV5jERR
0040	71 5a 69 63 39 34 54 77 48 4c 71 46 51 34 78 35	qZic94Tw HLqFQ4x5
0050	61 62 46 30 77 55 6e 59 73 46 2b 67 6c 44 47 4c	abF0wUnY sF+glDGL
0060	33 56 75 35 65 61 33 4d 44 59 77 49 70 63 32 44	3Vu5ea3M DYwIpc2D
0070	78 4c 44 4d 74 38 6b 2f 75 42 68 38 6a 48 6d 30	xLDMt8k/ uBh8jHm0
0080	63 66 54 63 69 35 6a 77 77 4c 2f 56 4c 6f 6c 41	cfTci5jw wL/VLo1A
0090	57 4c 6c 35 63 43 79 4e 6d 63 36 52 70 58 57 7a	WL15cCyN mc6RpXWz

Acknowledgment number (tcp.ack), 4 bytes
 Packets: 3083 · Displayed: 3083 (100.0%) | Profile: Default

WIRESHARK – FOLLOW TCP STREAM



WIRESHARK – FOLLOW TCP STREAM



NETWORKMINER - HOSTS

The screenshot displays the NetworkMiner 2.7.3 application window. The interface includes a menu bar (File, Tools, Help), a toolbar with 'Start' and 'Stop' buttons, and a dropdown menu to 'Select a network adapter in the list'. The main area is divided into two panes. The left pane, titled 'Anomalies', shows a list of hosts under the 'Hosts (179)' tab. The right pane, titled 'Case Panel', shows a table with 'Filename' and 'MD5' columns.

Hosts List:

- 192.168.0.1 (selected)
- 192.168.0.2 (Linux)
- 192.168.0.50
 - IP: 192.168.0.50
 - MAC: 00606E42B635
 - NIC Vendor: DAVICOM SEMICONDUCTOR, INC.
 - MAC Age: 1998-04-22
 - Hostname:
 - OS: Unknown
 - TTL: 1 (distance: 31)
 - Open TCP Ports:
 - Sent: 205 packets (15,400 Bytes), 0.00% cleartext (0 of 0 Bytes)
 - Received: 0 packets (0 Bytes), 0.00% cleartext (0 of 0 Bytes)
 - Incoming sessions: 0
 - Outgoing sessions: 0
 - Host Details
 - Queried DNS names : _ipps._tcp.local
 - UPnP field : HOST: 239.255.255.250 : 1900
 - UPnP field : MAN: "ssdp : discover"
 - UPnP field : M-SEARCH * HTTP/1.1
 - UPnP field : MX : 2
 - UPnP field : ST: roku : ecp
- 192.168.0.51 (Windows)
 - IP: 192.168.0.51
 - MAC: ECF4BB4FB245
 - NIC Vendor: Dell Inc.
 - MAC Age: 2013-11-08
 - Hostname:
 - OS: Windows
 - TTL: 64 (distance: 0)
 - Open TCP Ports:
 - Sent: 1738 packets (217,223 Bytes), 0.00% cleartext (0 of 0 Bytes)
 - Received: 2927 packets (3,797,852 Bytes), 0.00% cleartext (0 of 0 Bytes)
 - Incoming sessions: 0
 - Outgoing sessions: 6
 - Host Details
 - Queried DNS names : _ipps._tcp.local, safebrowsing.google.com, safebrowsing-cache.google.com
 - JA3 Hash 1 : aa7f5e2ada5d7bb8a7dceed01f5ffd7c
 - JA3 Hash 2 : 01f79a7537bf2cb8b8e8f450d291c632

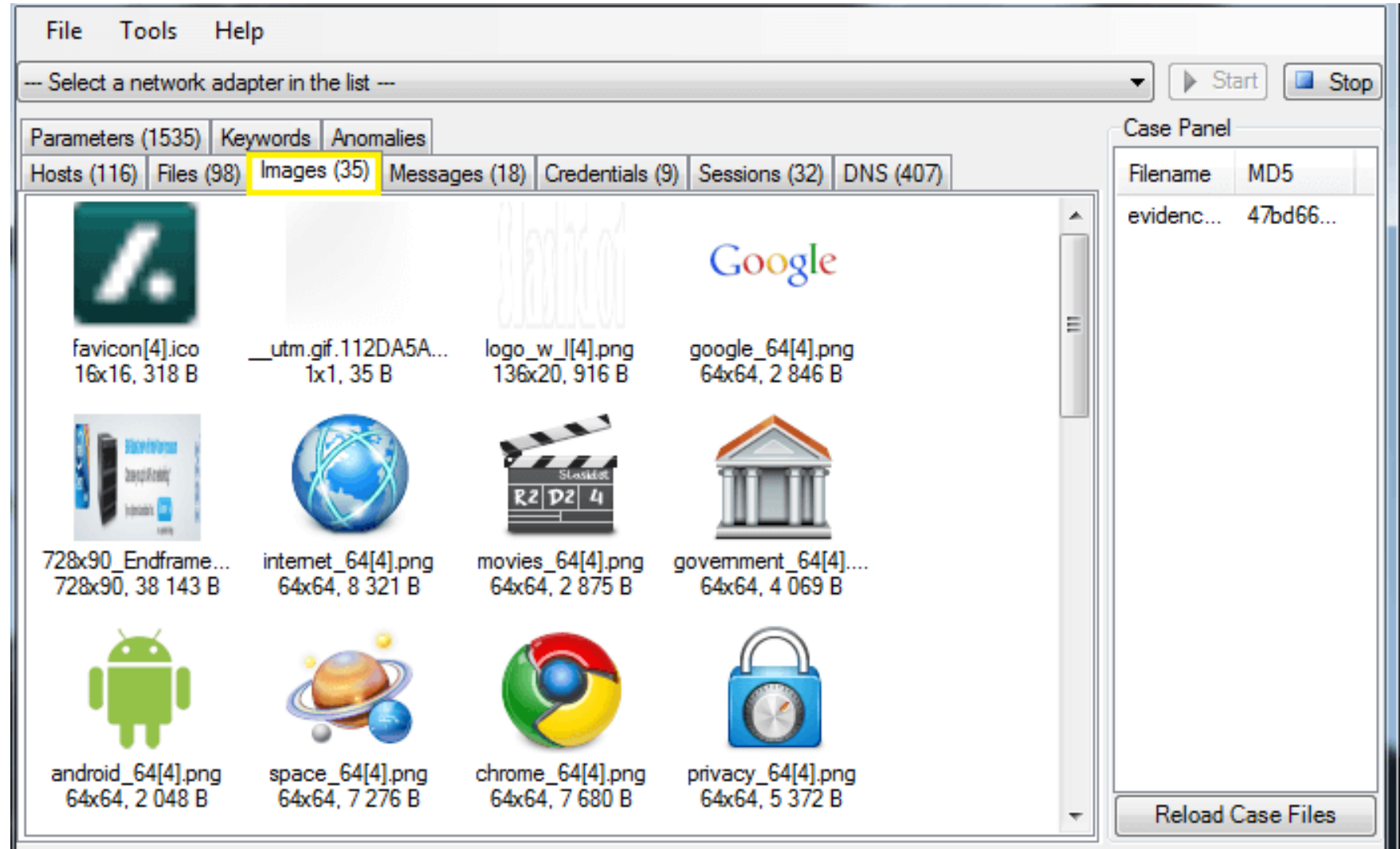
Case Panel:

Filename	MD5
snort.log....	f9b239b...

At the bottom, there is a 'Buffered Frames to Parse' field and a 'Reload Case Files' button.

NETWORKMINER

IMAGES TAB



NETWORKMINER

FILES TAB

File Tools Help

--- Select a network adapter in the list ---

Parameters (1535) Keywords Anomalies

Hosts (116) **Files (98)** Images (35) Messages (18) Credentials (9) Sessions (32) DNS (407)

Filter keyword: ☐ Case sensitive ExactPhrase Any column Clear Apply

Frame nr.	Filename	Extension	Size	Source host
111	index[4].html	html	84 143 B	216.34.181.45 [slashdot.org]
149	classic.css.FA90B794[4].css	css	86 482 B	184.30.179.172 [e872.g.akamaied
152	all-minified[4].js	js	263 843 B	184.30.179.172 [e872.g.akamaied
534	favicon[4].ico	ico	318 B	216.34.181.45 [slashdot.org]
554	jsct.1336660A[4].js	js	6 956 B	174.137.114.42 [jlinks.industrybrair
565	ga[4].js	js	26 707 B	74.125.224.105 [www-google-anal
585	__utm.gif.112DA5A[4].gif	gif	35 B	74.125.224.105 [www-google-anal
599	pg_index_p1_leader;p[4].js	js	6 192 B	74.125.224.123 [dart.l.doubleclick.
617	flashwrite_1_2[4].js	js	801 B	74.125.224.123 [dart.l.doubleclick.
621	logo w l[4].png	png	916 B	184.30.179.172 [e872.g.akamaied
643	noode 64[4].png	png	2 846 B	184.30.179.172 [e872.g.akamaied

Case Panel

Filename

evidenc...

NETWORKMINER

CREDENTIALS TAB

--- Select a network adapter in the list ---

Parameters (1535) Keywords Anomalies

Hosts (116) Files (98) Images (35) Messages (18) **Credentials (9)** Sessions (32) DNS (407)

☒ Show Cookies ☒ Show NTLM challenge-response ☐ Mask Passwords

Client	Server
192.168.30.108 [ann-laptop] [ann-laptop] (Windows)	74.125.224.123 [dart.l.doubleclick.net] [ad.doubleclick.net]
192.168.30.108 [ann-laptop] [ann-laptop] (Windows)	74.125.224.123 [ad.doubleclick.net]
192.168.30.108 [ann-laptop] [ann-laptop] (Windows)	74.125.224.122 [googleads.g.doubleclick.net]
192.168.30.108 [ann-laptop] [ann-laptop] (Windows)	74.125.224.122 [pagead.l.doubleclick.net] [googleads.g.doubleclick.net]
192.168.30.108 [ann-laptop] [ann-laptop] (Windows)	74.125.224.122 [googleads.g.doubleclick.net]
192.168.30.108 [ann-laptop] [ann-laptop] (Windows)	209.107.213.96 [a1294.v20.akamai.net] [b.scorecardresearch.com]
192.168.30.108 [ann-laptop] [ann-laptop] (Windows)	209.107.213.96 [b.scorecardresearch.com]
192.168.30.108 [ann-laptop] [ann-laptop] (Windows)	64.12.168.40 (Linux)
192.168.30.108 [ann-laptop] [ann-laptop] (Windows)	205.188.58.10 (Linux)

Case Panel

Filename

evidenc...



CHALLENGE A

- *access testasp.vulnweb.com*
 - start Wireshark
 - press login
 - enter Fictional username and password
 - find the password
 - save capture in a PCAP file



CHALLENGE B

- access testphp.vulnweb.com
 - start Wireshark
 - press Browse categories
 - press different links and images
 - save capture in a PCAP file
- *open saved PCAP files in NetworkMiner*
- *explore results*

THANK YOU

